

```
In [6]: import pandas as pd
sql=pd.read_csv(r'D:\Daily assignments\12th Feb\dataset_1_202602131113.csv')
```

```
In [7]: sql.head()
```

```
Out[7]:
```

	destination	passanger	weather	temperature	time	coupon	expiration	gender
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0	No Urgent Place	Alone	Sunny	55	2PM	Restaurant(<20)	1d	Femal
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1	No Urgent Place	Friend(s)	Sunny	80	10AM	Coffee House	2h	Femal
---	-----------------	-----------	-------	----	------	--------------	----	-------

2	No Urgent Place	Friend(s)	Sunny	80	10AM	Carry out & Take away	2h	Femal
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3	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House	2h	Femal
---	-----------------	-----------	-------	----	-----	--------------	----	-------

4	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House	1d	Femal
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5 rows × 27 columns



```
In [9]: sql.shape
```

```
Out[9]: (12684, 27)
```

```
In [10]: sql
```

Out[10]:

	destination	passanger	weather	temperature	time	coupon	expiration	g
0	No Urgent Place	Alone	Sunny	55	2PM	Restaurant(<20)	1d	F
1	No Urgent Place	Friend(s)	Sunny	80	10AM	Coffee House	2h	F
2	No Urgent Place	Friend(s)	Sunny	80	10AM	Carry out & Take away	2h	F
3	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House	2h	F
4	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House	1d	F
...	...	...	...	...	...	...	...	...
12679	Home	Partner	Rainy	55	6PM	Carry out & Take away	1d	
12680	Work	Alone	Rainy	55	7AM	Carry out & Take away	1d	
12681	Work	Alone	Snowy	30	7AM	Coffee House	1d	
12682	Work	Alone	Snowy	30	7AM	Bar	1d	
12683	Work	Alone	Sunny	80	7AM	Restaurant(20-50)	2h	

12684 rows × 27 columns



In [11]: `sql.columns`

Out[11]: Index(['destination', 'passanger', 'weather', 'temperature', 'time', 'coupon', 'expiration', 'gender', 'age', 'maritalStatus', 'has_children', 'education', 'occupation', 'income', 'car', 'Bar', 'CoffeeHouse', 'CarryAway', 'RestaurantLessThan20', 'Restaurant20To50', 'toCoupon_GEQ5min', 'toCoupon_GEQ15min', 'toCoupon_GEQ25min', 'direction_same', 'direction_opp', 'Y', 'row_count'], dtype='object')

In [12]: `#select weather,temperature from dataset_1;
sql[['weather', 'temperature']]`

Out[12]:

	weather	temperature
0	Sunny	55
1	Sunny	80
2	Sunny	80
3	Sunny	80
4	Sunny	80
...	...	...
12679	Rainy	55
12680	Rainy	55
12681	Snowy	30
12682	Snowy	30
12683	Sunny	80

12684 rows × 2 columns

In [13]: `sql[:11]`

Out[13]:

	destination	passanger	weather	temperature	time	coupon	expiration	genc
--	-------------	-----------	---------	-------------	------	--------	------------	------

0	No Urgent Place	Alone	Sunny	55	2PM	Restaurant(<20)	1d	Fem
1	No Urgent Place	Friend(s)	Sunny	80	10AM	Coffee House	2h	Fem
2	No Urgent Place	Friend(s)	Sunny	80	10AM	Carry out & Take away	2h	Fem
3	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House	2h	Fem
4	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House	1d	Fem
5	No Urgent Place	Friend(s)	Sunny	80	6PM	Restaurant(<20)	2h	Fem
6	No Urgent Place	Friend(s)	Sunny	55	2PM	Carry out & Take away	1d	Fem
7	No Urgent Place	Kid(s)	Sunny	80	10AM	Restaurant(<20)	2h	Fem
8	No Urgent Place	Kid(s)	Sunny	80	10AM	Carry out & Take away	2h	Fem
9	No Urgent Place	Kid(s)	Sunny	80	10AM	Bar	1d	Fem
10	No Urgent Place	Kid(s)	Sunny	80	2PM	Restaurant(<20)	1d	Fem

11 rows × 27 columns



In [14]: `#select *from dataset_1 LIMIT 10;`
`sql[0:11]`

Out[14]:

	destination	passanger	weather	temperature	time	coupon	expiration	gender
0	No Urgent Place	Alone	Sunny	55	2PM	Restaurant(<20)	1d	Female
1	No Urgent Place	Friend(s)	Sunny	80	10AM	Coffee House	2h	Female
2	No Urgent Place	Friend(s)	Sunny	80	10AM	Carry out & Take away	2h	Female
3	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House	2h	Female
4	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House	1d	Female
5	No Urgent Place	Friend(s)	Sunny	80	6PM	Restaurant(<20)	2h	Female
6	No Urgent Place	Friend(s)	Sunny	55	2PM	Carry out & Take away	1d	Female
7	No Urgent Place	Kid(s)	Sunny	80	10AM	Restaurant(<20)	2h	Female
8	No Urgent Place	Kid(s)	Sunny	80	10AM	Carry out & Take away	2h	Female
9	No Urgent Place	Kid(s)	Sunny	80	10AM	Bar	1d	Female
10	No Urgent Place	Kid(s)	Sunny	80	2PM	Restaurant(<20)	1d	Female

11 rows × 27 columns



In [17]: `sql['passanger'].nunique()`

Out[17]: 4

In [18]: `#select DISTINCT passanger from dataset_1;`
`sql['passanger'].unique()`

Out[18]: array(['Alone', 'Friend(s)', 'Kid(s)', 'Partner'], dtype=object)

In [19]: `#select *from dataset_1 WHERE destination='Home';`
`sql[sql['destination']=='Home']`

Out[19]:

	destination	passanger	weather	temperature	time	coupon	expiration	g
13	Home	Alone	Sunny	55	6PM	Bar	1d	F
14	Home	Alone	Sunny	55	6PM	Restaurant(20-50)	1d	F
15	Home	Alone	Sunny	80	6PM	Coffee House	2h	F
35	Home	Alone	Sunny	55	6PM	Bar	1d	
36	Home	Alone	Sunny	55	6PM	Restaurant(20-50)	1d	
...	...	...	...	...	...	...	...	
12675	Home	Alone	Snowy	30	10PM	Coffee House	2h	
12676	Home	Alone	Sunny	80	6PM	Restaurant(20-50)	1d	
12677	Home	Partner	Sunny	30	6PM	Restaurant(<20)	1d	
12678	Home	Partner	Sunny	30	10PM	Restaurant(<20)	2h	
12679	Home	Partner	Rainy	55	6PM	Carry out & Take away	1d	

3237 rows × 27 columns



In [24]: `#Select *FROM dataset_1 ORDER BY coupon;
sql.sort_values('coupon')`

Out[24]:

	destination	passanger	weather	temperature	time	coupon	expiration	g
11702	Home	Partner	Sunny	30	10PM	Bar	2h	F
9930	No Urgent Place	Alone	Snowy	30	2PM	Bar	1d	F
10632	Home	Alone	Rainy	55	6PM	Bar	1d	
7997	No Urgent Place	Friend(s)	Rainy	55	10PM	Bar	2h	
11166	Work	Alone	Snowy	30	7AM	Bar	1d	F
...	...	...	...	...	...	...	...	
10476	Home	Alone	Sunny	80	6PM	Restaurant(<20)	1d	F
5447	Home	Alone	Sunny	80	10PM	Restaurant(<20)	2h	F
10478	Home	Alone	Snowy	30	10PM	Restaurant(<20)	2h	F
5440	No Urgent Place	Alone	Sunny	80	2PM	Restaurant(<20)	2h	F
0	No Urgent Place	Alone	Sunny	55	2PM	Restaurant(<20)	1d	F

12684 rows × 27 columns



In [30]:

```
#Select destination as Destination FROM dataset_1;
sql.rename(columns={'destination': 'Destination'},inplace=True)
sql
```

Out[30]:

	Destination	passanger	weather	temperature	time	coupon	expiration	g
0	No Urgent Place	Alone	Sunny	55	2PM	Restaurant(<20)	1d	I
1	No Urgent Place	Friend(s)	Sunny	80	10AM	Coffee House	2h	I
2	No Urgent Place	Friend(s)	Sunny	80	10AM	Carry out & Take away	2h	I
3	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House	2h	I
4	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House	1d	I
...	...	...	...	...	...	...	...	
12679	Home	Partner	Rainy	55	6PM	Carry out & Take away	1d	
12680	Work	Alone	Rainy	55	7AM	Carry out & Take away	1d	
12681	Work	Alone	Snowy	30	7AM	Coffee House	1d	
12682	Work	Alone	Snowy	30	7AM	Bar	1d	
12683	Work	Alone	Sunny	80	7AM	Restaurant(20-50)	2h	

12684 rows × 27 columns



In [32]:

```
#SELECT occupation FROM dataset_1 GROUP BY occupation
sql.groupby('occupation').size().to_frame('Count').reset_index()
```


Out[32]:

	occupation	Count
0	Architecture & Engineering	175
1	Arts Design Entertainment Sports & Media	629
2	Building & Grounds Cleaning & Maintenance	44
3	Business & Financial	544
4	Community & Social Services	241
5	Computer & Mathematical	1408
6	Construction & Extraction	154
7	Education&Training&Library	943
8	Farming Fishing & Forestry	43
9	Food Preparation & Serving Related	298
10	Healthcare Practitioners & Technical	244
11	Healthcare Support	242
12	Installation Maintenance & Repair	133
13	Legal	219
14	Life Physical Social Science	170
15	Management	838
16	Office & Administrative Support	639
17	Personal Care & Service	175
18	Production Occupations	110
19	Protective Service	175
20	Retired	495
21	Sales & Related	1093
22	Student	1584
23	Transportation & Material Moving	218
24	Unemployed	1870

```
In [33]: # SELECT weather ,AVG(temperature) as avg_temp FROM dataset_1 GROUP BY weather;
sql.groupby('weather')['temperature'].mean().to_frame('avg_temp').reset_index()
```

Out[33]:

	weather	avg_temp
0	Rainy	55.000000
1	Snowy	30.000000
2	Sunny	68.946271

```
In [34]: #SELECT weather,COUNT(temperature) As count_temp FROM dataset_1 Group by weather
sql.groupby('weather')['temperature'].size().to_frame('count_temp').reset_index()
```

Out[34]:

	weather	count_temp
0	Rainy	1210
1	Snowy	1405
2	Sunny	10069

```
In [35]: sql.groupby('weather')['temperature'].nunique().to_frame('count_distinct_temp').res
```

Out[35]:

	weather	count_distinct_temp
0	Rainy	1
1	Snowy	1
2	Sunny	3

```
In [37]: sql.groupby('weather')['temperature'].sum().to_frame('sum_temp').reset_index()
```

Out[37]:

	weather	sum_temp
0	Rainy	66550
1	Snowy	42150
2	Sunny	694220

```
In [38]: sql.groupby('weather')['temperature'].min().to_frame('min_temp').reset_index()
```

Out[38]:

	weather	min_temp
0	Rainy	55
1	Snowy	30
2	Sunny	30

```
In [40]: sql.groupby('weather')['temperature'].max().to_frame('max_temp').reset_index()
```

```
Out[40]:
```

	weather	max_temp
0	Rainy	55
1	Snowy	30
2	Sunny	80

```
In [42]: sql.groupby('occupation').filter(lambda x: x['occupation'].iloc[0]=='Student').grou
```

```
Out[42]: occupation
Student    1584
dtype: int64
```

```
In [47]: pd.concat([sql, sql])['Destination'].drop_duplicates()
```

```
Out[47]: 0      No Urgent Place
13              Home
16              Work
Name: Destination, dtype: object
```

```
In [52]: sql[sql['passanger'] == 'Alone'][['Destination', 'passanger']]
```

```
Out[52]:
```

	Destination	passanger
0	No Urgent Place	Alone
13	Home	Alone
14	Home	Alone
15	Home	Alone
16	Work	Alone
...	...	...
12676	Home	Alone
12680	Work	Alone
12681	Work	Alone
12682	Work	Alone
12683	Work	Alone

7305 rows × 2 columns

```
In [53]: sql[sql['weather'].str.startswith('Sun')]
```

Out[53]:

	Destination	passanger	weather	temperature	time	coupon	expiration	g
0	No Urgent Place	Alone	Sunny	55	2PM	Restaurant(<20)	1d	l
1	No Urgent Place	Friend(s)	Sunny	80	10AM	Coffee House	2h	l
2	No Urgent Place	Friend(s)	Sunny	80	10AM	Carry out & Take away	2h	l
3	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House	2h	l
4	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House	1d	l
...	...	...	...	...	...	...	...	...
12673	Home	Alone	Sunny	30	6PM	Carry out & Take away	1d	
12676	Home	Alone	Sunny	80	6PM	Restaurant(20-50)	1d	
12677	Home	Partner	Sunny	30	6PM	Restaurant(<20)	1d	
12678	Home	Partner	Sunny	30	10PM	Restaurant(<20)	2h	
12683	Work	Alone	Sunny	80	7AM	Restaurant(20-50)	2h	

10069 rows × 27 columns



In [54]: `sql[(sql['temperature'] >= 29) & (sql['temperature'] <= 75)]['temperature'].unique()`

Out[54]: `array([55, 30])`

In [56]: `sql[sql['occupation'].isin(['Sales & Related', 'Management'])][['occupation']]`

Out[56]:

occupation	
193	Sales & Related
194	Sales & Related
195	Sales & Related
196	Sales & Related
197	Sales & Related
...	...
12679	Sales & Related
12680	Sales & Related
12681	Sales & Related
12682	Sales & Related
12683	Sales & Related

1931 rows × 1 columns

In []: